

5.5 Multiplying with pronumerals

In Algebra, the expression $3 \times n$ can be simplified by leaving out the multiplication sign.

$3 \times n$ is written as $3n$.



1 Simplify the following expressions.

- a** $4 \times a =$ 4a
- b** $5 \times y =$ _____
- c** $8 \times m =$ _____
- d** $a \times b =$ _____
- e** $k \times m =$ _____
- f** $3 \times X \times Y =$ _____
- g** $6 \times g \times h =$ _____
- h** $a \times b \times c =$ _____
- i** $k \times m \times n =$ _____

2 Write these expressions in expanded form, using multiplication signs.

- a** $5m =$ $5 \times m$
- b** $3k =$ _____
- c** $xy =$ _____
- d** $pqr =$ _____
- e** $9ab =$ _____
- f** $7pq =$ _____

When you multiply algebraic terms, multiply the numbers first.
Then multiply the pronumerals.
The pronumerals are usually written in alphabetical order.

EXAMPLES

$$\begin{aligned} 3 \times 2n & \\ = 3 \times 2 \times n & \\ = 6n & \end{aligned}$$

$$\begin{aligned} 4a \times 3b & \\ = 4 \times a \times 3 \times b & \\ = 4 \times 3 \times a \times b & \\ = 12ab & \end{aligned}$$

3 Simplify the following expressions.

- a** $2 \times 3y =$ _____
- b** $4 \times 5n =$ _____
- c** $6 \times 2a =$ _____
- d** $5 \times 8m =$ _____
- e** $9 \times 3k =$ _____
- f** $5y \times 4 =$ _____
- g** $7a \times 4 =$ _____
- h** $k \times 3 =$ _____
- i** $4m \times 1 =$ _____
- j** $4m \times 3n =$ _____
- k** $7a \times 4b =$ _____
- l** $8a \times 2b =$ _____
- m** $4k \times 5y =$ _____
- n** $5k \times 9m =$ _____
- o** $6p \times 7q =$ _____
- p** $3 \times 4a \times 5 =$ _____
- q** $7m \times n \times 5 =$ _____
- r** $2y \times k \times 6 =$ _____
- s** $4a \times 2b \times 3c =$ _____
- t** $4 \times 2b \times 7a =$ _____
- u** $6n \times 5k \times 2m =$ _____
- v** $3b \times a \times 5c =$ _____

5.5 Multiplying with pronumerals

CONTINUED

Is there another way to write $a \times a$?

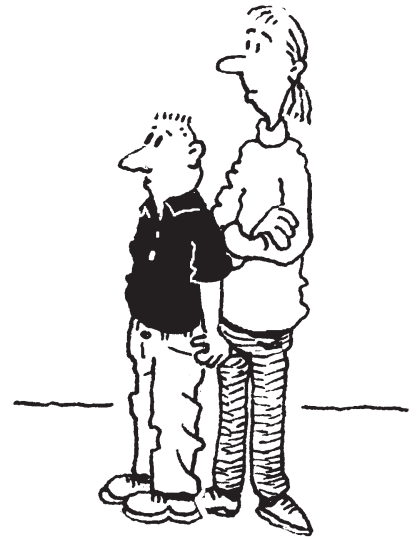
Index notation is a short way to write the number of times a pronumeral is multiplied by itself.

$a \times a$ becomes a^2 .

EXAMPLES

$$\begin{aligned} h \times h \times h \times h \\ = h^4 \end{aligned}$$

$$\begin{aligned} a \times a \times b \times b \times a \\ = a \times a \times a \times b \times b \\ = a^3 b^2 \end{aligned}$$



4 Write using index notation.

a $n \times n \times n \times n =$ _____

c $m \times m \times m \times m \times m =$ _____

e $a \times a \times b \times b \times b =$ _____

g $p \times q \times q =$ _____

i $b \times c \times c \times b =$ _____

k $a \times b \times a \times a \times b =$ _____

b $a \times a \times a =$ _____

d $y \times y =$ _____

f $k \times m \times k \times m =$ _____

h $a \times b \times a =$ _____

j $k \times k \times y \times y \times k =$ _____

l $k \times n \times n \times k \times n =$ _____

5 Simplify the following expressions.

a $9 \times m \times m \times m =$ _____

c $8a \times 2a =$ _____

e $d \times 3d =$ _____

g $5 \times n \times 2n =$ _____

i $6m \times 2m \times 4m =$ _____

k $4p \times 2p \times p \times p =$ _____

m $3k \times k \times y =$ _____

o $3m \times 2n \times 4m =$ _____

q $a \times ab \times b =$ _____

s $3ac \times 2ac =$ _____

b $5 \times a \times 4 \times a =$ _____

d $7y \times 3y =$ _____

f $5m \times 2m \times m =$ _____

h $3a \times a \times 4a =$ _____

j $2k \times 4k \times 3 =$ _____

l $8m \times 2 \times 5m =$ _____

n $5m \times a \times m =$ _____

p $a \times b \times 2a =$ _____

r $2k \times 3y \times 5k =$ _____

t $4m \times 3n \times mn =$ _____